NATIONAL MATHEMATICS CONTEST
O-LEVEL PAPER.

SATURDAY 28TH MAY, 2022 9:00 AM - 12:15 PM

INSTRUCTIONS TO CANDIDATES AND SUPERVISORS:

(a) *This competition is conducted on the assumption that proper security is maintained. UMS reserves the right, should there be evidence or suspicion of any malpractice, to reject scripts from a particular school/institution.*

(b) *There are* **two** *sections and each carries* 50 *marks. Attempt all questions in exactly three hours fifteen minutes (*$9:00am$ *to* $12:15pm$*).*

(c) *Indicate your names, birth date, gender, class, school and district* ***on all*** *your answer sheets.*

(d) *Marks will be awarded for only answers for which a clear and logical layout of the working is provided. A correct solution, poorly presented, will not earn full marks.*

(e) *Answer scripts of only registered students should be returned to the contest coordinator, latest Monday 30th May, 2022.*

(f) *ALL participants who score a pass mark are cordially invited to the certificate and prize giving ceremony on Saturday* 23*rd July,* 2022 *at 8:00am at Makerere University, Kampala.*

SECTION A : 5 marks each

1. The sequence $a_1, a_2, a_3, \ldots$ starts with $a_1 = 60$. To work out a_{n+1} for $n \geq 1$ you add 1 to the digit sum of a_n and square the result e.g

$$a_2 = (6 + 0 + 1)^2 = 49.$$

Find the value of the term a_{2022}.

2. If

$$\log_x 64 + \log_{y^2} 16 = \frac{5}{3}$$

and

$$\log_y 64 + \log_{x^2} 16 = 1$$

for all positive numbers x and y, what is the value of $\log_2(xy)$?

3. A function f satisfies $f(xy) = \dfrac{f(x)}{y^2}$ for all positive real numbers x and y. Given that $f(25) = 80$, Find the value of $f(2020)$.

4. Given that $xy > 0$ and that $x - y = xy$. Find the value of

$$\frac{120}{x} - \frac{120}{y} + 2022.$$

5. John invested in a business and lost 20% during the first year of the business. The following year he made a gain of 25%. What is his net gain over the two years?

6. If $f(\sin x) = \sin 3x$. Evaluate $f(\cos x)$.

7. Find the length of a side of a square if a circle that passes through one corner of the square and is tangent internally to two sides of the square has radius 10.

8. Suppose a bag contains six letters of the word "GOOGOO". If you take one letter out of the bag at a time and line them up from left to right. What is the probability that you will spell the word "GOOGOO"?

9. What is the 2022nd term in the sequence?

$$\frac{1}{1}, \frac{2}{1}, \frac{1}{2}, \frac{3}{1}, \frac{2}{2}, \frac{1}{3}, \frac{4}{1}, \frac{3}{2}, \frac{2}{3}, \frac{1}{4}, \frac{5}{1}, \ \ldots$$

10. Suppose it takes h minutes to fill a tank using inlet A and c minutes to fill the same tank using inlet B. Starting with an empty tank, inlet A is turned on and after one minute inlet B is also turned on. How long will it take to fill the tank.[Give your answer in terms of h and c]

SECTION B : 10 **marks each**

11. Opio and Peter ran a race, during which they both ran at constant speeds. Opio began the race $30m$ ahead of Peter. After 3 minutes, Peter was 18 m ahead of Opio. Peter won the race exactly 7 minutes after it began. How far from the finish line was Opio when Peter won?

12. Given that the equation $\left(\frac{1}{4}\right)^x + \left(\frac{1}{2}\right)^{x-1} + b = 0$ has a positive solution. What is the interval in which the real number b should be located?

13. A box contains colored balls, each marked with a positive integer. Half of the balls are blue and the others are gold. One fifth of the balls are marked with prime numbers. Four dozen are marked with numbers that are not prime. One third of the blue balls are marked with prime numbers. How many;

    (a) of the gold balls are marked with numbers that are not prime?

    (b) balls are not blue or are marked with numbers that are not prime or are both not blue and marked with a number that is not prime?

14. If the following pattern continues. Name the row on which the number 289 will appear. [Explain your answer]

$$\begin{array}{ccccccccccc}
 & & & & 1 & & & & & & \\
 & & & 3 & & & 5 & & & & \\
 & & 7 & & & 9 & & & 11 & & \\
13 & & & 15 & & & 17 & & & 19 &
\end{array}$$

15. A goat is tied to the corner of a building whose base is $8ft$ by $10ft$ with a twelve feet rope. How many square feet of grass can the goat reach to eat.[Ignore the dimensions of the goat].

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SECTION A : 5 marks each

1. **Solution :**

$$\begin{aligned}
a_2 &= (6+0+1)^2 = 49\\
a_3 &= (4+9+1)^2 = (14)^2 = 196\\
a_4 &= (1+9+6+1)^2 = 289\\
a_5 &= (2+8+9+1)^2 = 400\\
a_6 &= (4+0+0+1)^2 = 25\\
a_7 &= (2+5+1)^2 = 64\\
a_8 &= (6+4+1)^2 = 121\\
a_9 &= (1+2+1+1)^2 = 25
\end{aligned}$$

The process will repeat after three steps. This means that

$$a_6 = a_9 = a_{12} = 25.$$

Since 2022 is a multiple of 3 *then* $a_{2022} = 25$. ■

2. **Solution :** *Convert the logarithms to base two.*

$$\frac{\log_2 64}{\log_2 x} + \frac{\log_2 16}{\log_2 x^2} = \frac{5}{3}$$

$$\frac{\log_2 64}{\log_2 y} + \frac{\log_2 16}{\log_2 y^2} = 1$$

On reducing and evaluating the logarithms we get,

$$\frac{6}{\log_2 x} + \frac{2}{\log_2 y} = \frac{5}{3}$$

$$\frac{6}{\log_2 y} + \frac{2}{\log_2 x} = 1$$

Letting

$$\begin{aligned} \log_2 x &= a \\ \log_2 y &= b \end{aligned}$$

and solving the above equation we have, $a = 4,\ b = 12$ *and therefore*

$$\log_2(xy) = \log_2 x + \log_2 y = 16.$$

■

3. **Solution :**

$$\begin{aligned} f(xy) &= \frac{f(x)}{y^2} \\ f(25) &= 80 \end{aligned}$$

Therefore

$$xy = 25$$

Substituting $x = \dfrac{25}{y}$ *into* $\dfrac{f(x)}{y^2} = 80$ *we get*

$$f(x) = \frac{80 \times 625}{x^2}$$

For $f(2020)$ *we have* $xy = 2020 \Rightarrow x = \dfrac{2020}{y}$ *and* $y = \dfrac{2020}{x}$ *and*

$$f(2020) = \frac{f(x)}{y^2} = \frac{80 \times 65}{x^2} \times \frac{1}{\dfrac{(2020)^2}{x^2}} = \frac{500}{40804} = \frac{125}{10201} = 0.012253.$$

■

4. **Solution :**

$$120\left(\frac{1}{x} - \frac{1}{y}\right) + 2022 = 120\left(\frac{-(y-x)}{xy}\right) + 2022 = -120 + 2022 = 1902.$$

■

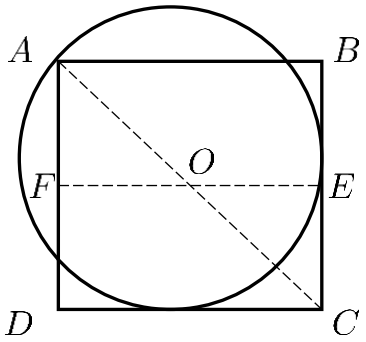
5. **Solution :** *Let* x *be the original investment into the business in* 2020*, After the first year (2020) the amount invested would be* $0.8x$*, after the second year the investment will be increased by* 25% *thus become* $1.25 \times 0.8x = x$ *implying that the net gain is zero.* ■

6. **Solution :** *Recall* $\cos 30^\circ = \sin 60^\circ$ *so*

$$f(\cos 30^\circ) = f(\sin 60^\circ) = \sin(60^\circ \times 3) = \sin 180^\circ = 0.$$

■

7. **Solution :** *Let $ABCD$ be the square with A on the circle as shown.*



Let O be the center of the circle, let BC be tangent to the circle at E, Draw the diagonal AC passing through O. Draw OE, extend EO intersecting AD at F, we have $OE = OA$ =10. AFO is an isosceles right angle therefore

$$OF = \frac{10}{\sqrt{2}} = 5\sqrt{2}$$

Therefore,

$$\begin{aligned} EF &= 10 + OF \\ &= 10 + 5\sqrt{2} \end{aligned}$$

■

8. **Solution :** *There are* 6! *ways of lining up letters in the word "GOOGOO" without regard of which letters are where. There are:*

- *2 ways of choosing the first letter as G,*
- *4 ways of choosing the second letter as "O",*
- *3 ways of choosing the 3rd letter as "O" and*
- *1 way of choosing the fourth letter as G since one G is already taken for beginning,*
- *2 ways of choosing an "O" since there are only two left and*
- *1 way for the last letter "O".*

Therefore

$$P(GOOGOO) = \frac{2 \times 4 \times 3 \times 1 \times 2 \times 1}{6!} = \frac{1}{15}$$

■

9. **Solution :** *Given the sequence*

$$\frac{1}{1}, \frac{2}{1}, \frac{1}{2}, \frac{3}{1}, \frac{2}{2}, \frac{1}{3}, \frac{4}{1}, \frac{3}{2}, \frac{2}{3}, \frac{1}{4}, \frac{5}{1}, \quad \cdots$$

We note that the first three terms have the numerator increasing from 1 *to* 2 *and back to* 1 *while denominator increases to* 2*, the next three terms have the numerator decrease from* 3 *to* 1 *while denominator increase to* 3*, the next four terms have the numerator decrease from* 4 *to* 1 *while denominator increase to* 4*, the next five terms have the numerator decrease from* 5 *to* 1 *while denominator increase to* 5*, and so on.*

We note that the number of terms will be in the format $3, 3, 4, 5, 6, 7, 8, 9, \ldots$. *Now given that* $3+3 = 1+2+3$ *then the number of terms of the sequence where the numerator of the terms of* n *decreasing from* n *to* 1 *while denominator increasing from* 1 *to* n *will be of the form*

$$3 = 1+2,\ 3,\ 4,\ 5,\ 6,\ 7,\ 8,\ 9,\ \ldots$$

therefore the number n *closest to* 2022 *such that* $\dfrac{n(n+1)}{2} = 2022$ *is* 63 *that is*

$$\frac{63(64)}{2} = 2016$$

and the next terms will be

$$\frac{64}{1},\ \frac{63}{2},\ \frac{62}{3},\ \frac{61}{4},\ \frac{60}{5},\ \frac{59}{6},\ \ldots$$

Thus the 2022nd *term is*

$$\frac{59}{6}$$

■

10. **Solution :** *After* 1 *minute, there is only* $\left(1-\dfrac{1}{h}\right)$ *of the total space in the tank left to be filled.*

Let t *be the total time it takes to fill the tank when both inlets are open, then*

$$\frac{1}{h}+\frac{1}{c} = \frac{1}{t} \quad \Rightarrow \quad t = \frac{hc}{h+c}$$

The total time it takes for the tank to fill is

$$\begin{aligned} 1\ \textit{minute}\ + \left(1-\frac{1}{h}\right)\left(\frac{hc}{h+c}\right) &= 1+\left(1-\frac{1}{n}\right)\left(\frac{hc}{h+c}\right) \\ &= \frac{h+hc}{h+c} \\ &= \frac{h(1+c)}{h+c} \end{aligned}$$

■

11. **Solution :** *Over the* 3 *min of the race, Peter ran* 48 *m further than Opio. Why?*

We note that at 0 *mins; Opio was at* $30m$ *mark. If Opio ran* d *m in the* 3 *minutes then he will be at* $(d+30)m$ *mark after* 3 *minutes. Since Peter is* $18m$ *ahead of Opio after* 3 *mins then Peter is at*

$$(d+30+18) = d+48$$

Therefore, In 3 *minutes Peter ran* $(d+48)m$ *which is* $48m$ *further than Opio.*

Since each runs at a constant speed then Peter runs $\frac{48}{3} = 16m/min$ *faster than Opio.*

As Peter finishes the race after 7 *minutes then Peter runs for another* 4 *mins.*

Over the 4 *mins, he runs* $4 \times 16 = 64$ *m further than Opio. After* 3 *mins Peter runs* 64 *meters further than Opio. So Opio is* 64 *m from the finish line.* ■

12. **Solution :**

$$\begin{aligned}\left(\frac{1}{4}\right)^x + \left(\frac{1}{2}\right)^{x-1} + b &= 0\\ \left(\left(\frac{1}{2}\right)^x\right)^2 + 2\left(\frac{1}{2}\right)^x + b &= 0\end{aligned}$$

Let

$$y = \left(\frac{1}{2}\right)^x$$

such that

$$y^2 + 2y + b = 0$$

to have

$$\begin{aligned}y &= \left(\frac{1}{2}\right)^x = -1 \pm \sqrt{1-b}\\ x &= \frac{\ln\left(\sqrt{1-b}-1\right)}{-\ln 2}\end{aligned}$$

We need $\ln\left(\sqrt{1-b}-1\right) < 0$ *for a positive solution.*

$$0 < \sqrt{1-b} - 1 < 1 \quad \Rightarrow \quad 1 < \sqrt{1-b} < 2 \quad \Rightarrow \quad -3 < b < 0$$

■

13. **Solution :** *Four dozen is* $1 - \frac{1}{5} = \frac{4}{5}$ *of the total no of balls. The total number of balls is*

$$48 \div \frac{4}{5} = 60$$

- *There are* $60 - 48 = 12$ *balls marked with prime numbers and* 48 *balls marked with non prime numbers.*
- *There are* 30 *blue balls and* 30 *gold balls*
- *Further there are* $\frac{1}{3} \times 30 = 10$ *blue balls marked with prime numbers.*
- *There are* $12 - 10 = 2$ *gold balls marked with prime numbers*
- *Therefore there are* 28 *gold balls marked with non prime number*

	Blue	*Gold*	*Total*
Prime	10	2	12
Non-prime	20	28	48
Total	30	30	60

$$60 - 10 \;=\; 50$$

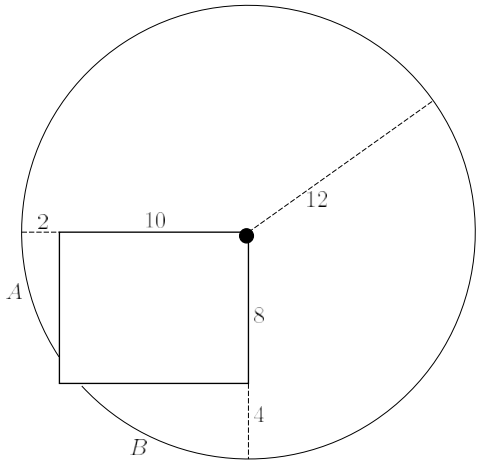
■

14. **Solution :**

$$\begin{array}{ccccccccccc}
 & & & & 1 & & & & & & \\
 & & & 3 & & 5 & & & & & \\
 & & 7 & & 9 & & 11 & & & & \\
 & 13 & & 15 & & 17 & & 19 & & & \\
21 & & 23 & & 25 & & 27 & & 29 & & \\
 & 31 & & 33 & & 35 & & 37 & & 39 & & 41 & \\
43 & & 45 & & 47 & & 49 & & 51 & & 53 & & 55
\end{array}$$

The number $289 \;=\; (17)^2 \Rightarrow$ *it appears in the* 17th *row and in the* 9th *element since the* 1st *row has* 1, 3rd *row has* $3^2 = 9$, 5th *row has* $5^2 = 25$ *in the middle, . . .* ■

15. **Solution :**



Total area that the goat reach is;

- $\frac{3}{4}$ *of the circle whose radius is* $12\ ft$*; i.e*

$$\frac{3}{4}\big(\pi r^2\big) \;=\; \frac{3}{4}\big(144\big)\pi$$

and

- $\frac{1}{4}$ *of the circular part*

$$A \;=\; \frac{1}{4}(\pi \cdot 2^2) = \pi$$

and

- $\frac{1}{4}$ *of the circular part*

$$B \;=\; \frac{1}{4}(\pi \cdot 4^2) = 4\pi$$

Total area reached is

$$\frac{3}{4}\big(144\big)\pi + \pi + 4\pi \;= 113\pi.$$

■